Night Logger, Lab report

Zurich University of Applied Sciences
School of Engineering

Author Gabriel Gillmann

Supervisors Dr. Y. Lukic

Eingereicht am: 12.04.2026

1 Problem

Patients with chronic cough, asthma, or sleep apnea are frequently unaware of how often they cough or snore during the night [1]. Existing solutions either require clinical grade equipment or upload audio to cloud servers, raising privacy concerns [2, 3]. NightLogger provides a local, on device sleep sound monitor that runs passively overnight on a smartphone placed on the bedside table. In the morning it presents an annotated summary of detected respiratory events, without ever transmitting audio off the device. Target users: patients managing a chronic respiratory condition, or anyone curious about their sleep quality, who want a lightweight, privacy respecting logging tool.

2 Related work

Prior work on passive acoustic monitoring established that coughs and snores have sufficiently distinct spectro-temporal signatures for machine classification [4]. Commercial apps such as SnoreLab record raw audio and stores it in the cloud [5]. NightLogger takes the opposite approach: no audio ever leaves the device.

3 Model choice

Apple SNClassifySoundRequest, a CoreML-backed classifier trained on millions of clips across 300+ classes, including cough, snoring, sneeze, speech, gasping, and wheezing and runs entirely via the Neural Engine. The model acts as a recognizer: it takes a 1-second audio window (50 % overlap) and returns ranked class probabilities [6].

4 Interaction design

Recording screen: one large button starts the session. A pulsing ring animation and live event feed confirm the app is listening. For overnight use, Auto-Lock should be disabled and the device kept on charge.

Detected sound classes: cough, snoring, sneeze, sleep talking, gasping and unknown.

Dashboard: stat cards per class with medical notes, a bar chart breakdown, and an hourly timeline histogram.

UI Policy, uncertainty handling:

- Confidence $< 45\%$ $\rightarrow$ silently discarded.
- 45–69% (medium) $\rightarrow$ 15-second audio clip saved. User listens back, then confirms, reclassifies, or dismisses.
- $\geq 70\%$ (high) $\rightarrow$ counted immediately. Sleep talking clips always saved for morning playback.

For **Unknown** events a full class grid lets the user relabel the clip, correcting misclassifications rather than simply accepting or discarding them. This is the primary UI fallback strategy for model uncertainty.

Sleep talking: when speech is detected at night a 15-second clip is always saved regardless of confidence, so the user can listen back to what they said in the morning.

5 Evaluation

Controlled test: 30 manually produced coughs, snores, gasps, sleep talking, and sneezes played through a speaker at 0.5 m. Results:

- **Cough detection rate:** 28/30 (93.3 %).
- **Snore detection rate:** 24/30 (80 %).
- **Sneeze detection rate:** 20/30 (66.6 %).
- **Sleep talking detection rate:** 18/30 (60.0 %).
- **Gasping detection rate:** 4/30 (13.3 %).

Deployment constraints: Auto-Lock must be disabled for overnight use; device should be kept on charge. Latency: 1 s per window. Memory: 15-second rolling buffer (5 MB). Privacy: no audio transmitted; only timestamped metadata stored locally as JSON.

6 Reflection

What worked well: the confidence-tiered UI policy proved effective, the morning triage workflow felt natural but took a few minutes for a full night of data. Being able to hear a flagged sound before deciding whether to count it makes triage genuinely useful rather than arbitrary.

Limitations: the classifier was not trained specifically on sleep audio. quiet sounds, especially gasping were sometimes missed. Requiring Auto Lock to be disabled is a friction point for casual users.

Overall, NightLogger shows that a focused, privacy first interaction design powered by a single on device model with a well designed uncertainty policy can produce a useful tool.

Bibliography

[1] F. Barata et al. "Automatic Recognition, Segmentation, and Sex Assignment of Nocturnal Asthmatic Coughs and Cough Epochs in Smartphone Audio Recordings: Observational Field Study". In: *Journal of Medical Internet Research* 22.7 (2020), e18082. DOI: 10.2196/18082. URL: https://www.jmir.org/2020/7/e18082.

[2] Marina Eni, Valeria Mordoh, and Yaniv Zigel. "Cough detection using a non-contact microphone: A nocturnal cough study". In: *PLOS ONE* 17.1 (2022), e0262240. DOI: 10.1371/journal.pone.0262240. URL: https://doi.org/10.1371/journal.pone.0262240.

[3] C. L. V. Sivakumar, Varda Mone, and Rakhmanov Abdumukhtor. "Addressing privacy concerns with wearable health monitoring technology". In: *WIREs Data Mining and Knowledge Discovery* 14.3 (2024), e1535. DOI: https://doi.org/10.1002/widm.1535. eprint: https://wires.onlinelibrary.wiley.com/doi/pdf/10.1002/widm.1535. URL: https://wires.onlinelibrary.wiley.com/doi/abs/10.1002/widm.1535.

[4] T. Drugman et al. "Objective study of sensor relevance for automatic cough detection". In: *IEEE Journal of Biomedical and Health Informatics* 17.3 (May 2013), pp. 699–707. DOI: 10.1109/JBHI.2013.2239303.

[5] Reviva Softworks Ltd. *SnoreLab's Privacy Notice*. SnoreLab. Feb. 3, 2026. URL: https://www.snorelab.com/snorelab-app-privacy-policy (visited on 04/12/2026).

[6] Apple Inc. *SNClassifySoundRequest | Apple Developer Documentation*. https://developer.apple.com/documentation/soundanalysis/snclassifysoundrequest. Abgerufen am: 12. April 2026. 2026.